

# OPTIMAL SCORING STRATEGIES FOR THE LIZARD CULT IN *Root*

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We propose a discrete-time Markov chain framework to model scoring using gardens for the Lizard Cult faction in the asymmetric board game *Root*. The Cult's ability to score from their gardens is affected three different processes with stochasticity introduced by their opponents: suited card draws from a shared deck, the dynamic Outcast suit determined by recently discarded cards, and forcible removal of their gardens. Unlike other factions, the Cult chooses when it will score from its gardens, which allows for significant strategy-based variation. We use Monte Carlo simulation to visualize scoring trajectories and determine which parameters are most key.

## 1 INTRODUCTION

*Root* is a board game with a high degree of asymmetry: each player chooses a unique faction out of 10. Every faction has unique actions and their own ways to score and 'win the war'. For the Lizard Cult faction, the primary victory mechanism involves building *gardens* and performing the Score ritual [1] (Appendix F). Most other factions have automatic or incidental automatic scoring, but the Cult chooses when it scores points (at the expense of its resources), which creates an optimization problem. Three sources of randomness complicate this optimization:

1. **Card draws:** Rituals require suited cards drawn from a shared 54-card deck. The Cult's "Hatred of Birds" rule means the 18 Bird-suit cards cannot be used for scoring.
2. **Garden dynamics:** Gardens grow in number through player actions (the Build ritual, the Cult's Conspiracies) but are likely to be destroyed by opponents through combat and special abilities.
3. **Hand composition:** The random distribution of which suits of cards will be drawn cards constrains the suits which can be scored each turn.

### 1.1 Game Mechanics

The Cult's faction board has three garden tracks (Fox, Rabbit, Mouse), each with capacity 5. Let  $\mathbf{G} = (G_F, G_R, G_M) \in \{0, \dots, 5\}^3$  denote gardens on the map. The Cult starts with 1 garden (of chosen suit) on the map [1].

**Scoring.** On the Cult's turn, performing the Score ritual for suit  $s$  yields victory points based on garden count, as per the faction board:

$$S(g) = \begin{cases} 0 & g \in \{0, 1\} \\ 2 & g \in \{2, 3\} \\ 3 & g = 4 \\ 4 & g = 5 \end{cases} \quad (1)$$

**Card draw.** The Cult draws  $n(\mathbf{G}) = 1 + \text{Bonus}(\mathbf{G})$  cards per turn, where:

$$\text{Bonus}(\mathbf{G}) = \sum_s \mathbb{1}[G_s \geq 2] \quad (2)$$

counts draw bonuses (one per suit with  $\geq 2$  gardens). After drawing, the Cult must discard down to a hand limit of 5 cards. Since discards are chosen by the player, this allows preferential retention of suited cards for scoring on subsequent turns.

**Scoring constraints.** Each suit can be scored at most once per turn, but the Cult can score multiple suits (up to 3) in a single turn. Scoring requires spending a suited card.

**Victory.** The first player to 30 VP wins immediately.

### 1.2 Threshold Policies

We formalize the Cult's choice as to whether they should score or wait as a threshold policy.

**Definition 1** (Threshold Policy). Our threshold policy  $\pi_\theta$  with parameter  $\theta \in \{1, \dots, 5\}$  is:

- **Score** suit  $s$  if  $G_s \geq \theta$  and a suit- $s$  card is available
- **Build**<sup>1</sup> otherwise

with  $\theta^*$  minimizing expected turns to victory.

<sup>1</sup>The mechanisms that 'Build' encompasses are vast, and we cannot model them all here. In general, the Cult should use the Sacrifice ritual on Bird cards, and complete Recruit rituals. These will complement the Build ritual and Sanctify conspiracy, which allow the Cult to get more gardens on the map.

**Definition 2** (Victory Time). The victory time under policy  $\pi_\theta$  is the hitting time:

$$\tau(\theta) = \inf\{t \geq 0 : V_t \geq 30\} \quad (3)$$

which gives  $\theta^* = \arg \min_\theta \mathbb{E}[\tau(\theta)]$ .

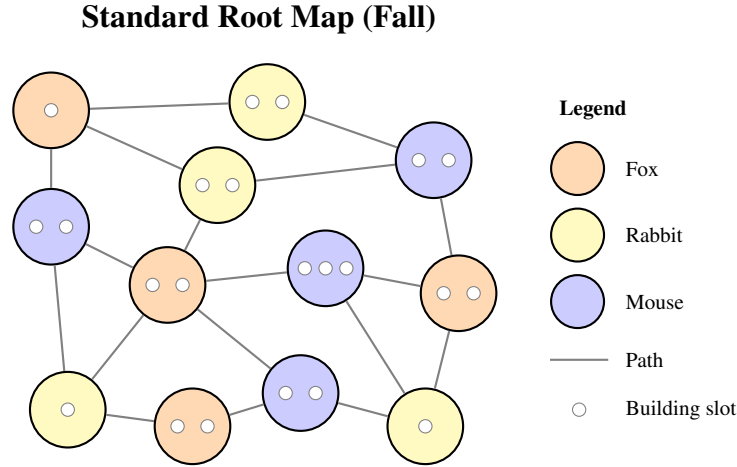


Figure 1: The standard (Fall) map). Clearings are colored by suit; white circles indicate all possible building slots where the Lizard Cult can place gardens (note: in some games, certain slots have Ruins and are unusable) [1].

## 2 MARKOV CHAIN MODEL

We define a finite-state Markov chain that captures the essence of the relevant strategic working of the Cult.

### 2.1 Reduced State Space

Tracking the full game state is both overly complicated and quite difficult; the state space becomes intractably large. We instead define a reduced state space for the key aspects of strategy.

**Definition 3** (Reduced State Space). The state on turn  $t$  is given by:

$$\mathbf{X}_t = (\mathbf{G}_t, V_t, \mathbf{h}_t) \in \mathcal{S} \quad (4)$$

where:

- $\mathbf{G}_t = (G_F^t, G_R^t, G_M^t) \in \{0, 1, 2, 3, 4, 5\}^3$ : garden counts per suit
- $V_t \in \{0, 1, \dots, 29\}$ : victory points (states with  $V \geq 30$  are absorbing)
- $\mathbf{h}_t = (h_F^t, h_R^t, h_M^t) \in \{0, 1\}^3$ : indicators where  $H_s = 1$  iff at least one suit- $s$  card is in hand

The garden count is given by  $|\mathbf{G}| = G_F + G_R + G_M$ . The state space has  $|\mathcal{S}| = 6^3 \times 30 \times 2^3 = 51,840$  transient states, and one absorbing “victory” state.

The hand indicator  $\mathbf{h}$  captures whether scoring suit  $s$  is possible this turn, instead of considering how many suited cards are available. This does not affect the analysis of scoring, as the Cult can only score each suit once per turn. The other approximation is a fresh-deck approximation: in the model, we draw from the full 54-card deck each turn. In particular, draws from the deck are independent of existing hand composition. Although this is a simplification, it is

both necessary for computation (otherwise, there are 41,743 deck states) and reasonable since the Cult only cares about suit distribution (and not specific cards).

### 2.2 Turn Structure and Transition Kernel

We adapt *Root*’s turn structure language of *Birdsong*, *Daylight*, and *Evening*. In *Birdsong*, the Cult can Sanctify new gardens into play; in *Daylight*, they can perform rituals, including Build and Score. Logically, we assume that players will Build before they Score, allowing us to simplify. They then Draw, and pass to other players. We model this in four phases: growth, scoring, card draw, then attrition.

$$(\mathbf{G}, V, \mathbf{h}) \xrightarrow{1: \text{Growth}} \mathbf{G}' \xrightarrow{2: \text{Score}} (V', \mathbf{h}') \xrightarrow{3: \text{Draw}} \mathbf{h}'' \xrightarrow{4: \text{Attrition}} (\mathbf{G}'', \mathbf{h}''')$$

The next-turn state is  $\mathbf{X}_{t+1} = (\mathbf{G}'', V', \mathbf{h}''')$ .

#### 2.2.1 Phase 1: Growth of Gardens

Gardens can grow through Build rituals and the Sanctify conspiracy; since this depends highly on player behaviour, luck, and other factors, we choose to model this as independent Bernoulli growth using an aggregate parameter  $\mu$ .

**Definition 4** (Growth Transition). Each suit’s garden count increases by 1 with probability  $\mu$ , capped at 5:

$$\begin{aligned} \mathbb{P}(G'_s = g+1 \mid G_s = g) &= \mu \cdot \mathbb{1}[g < 5] \\ \mathbb{P}(G'_s = g \mid G_s = g) &= 1 - \mu \cdot \mathbb{1}[g < 5] \end{aligned} \quad (5)$$

Suits grow independently, so:

$$\mathbb{P}(\mathbf{G}' \mid \mathbf{G}) = \prod_{s \in \{F, R, M\}} \mathbb{P}(G'_s \mid G_s) \quad (6)$$

*Remark 5* (Interpretation of the aggregate  $\mu$ ). In practice,  $\mu \in [0.2, 0.5]$  depending on luck and opponent pressure (for example, the Cult may need to defend gardens instead of building new ones). See Section 6 for sensitivity analysis.

### 2.2.2 Phase 2: Scoring

Under a threshold policy  $\pi_\theta$ , the Cult scores all eligible suits.

*Definition 6* (Scoring Transition). Given threshold  $\theta$ , garden configuration  $\mathbf{G}'$ , and hand  $\mathbf{h}$ :

- For each suit  $s$ : if  $G'_s \geq \theta$  and  $h_s = 1$ , score  $S(G'_s)$  points and set  $h'_s = 0$
- Otherwise:  $h'_s = h_s$

The VP update is:

$$V' = \min \left( V + \sum_s S(G'_s) \cdot \mathbb{1}[G'_s \geq \theta] \cdot h'_s, 30 \right) \quad (7)$$

The hand update is:

$$h'_s = h_s \cdot (1 - \mathbb{1}[G'_s \geq \theta]) \quad (8)$$

### 2.2.3 Phase 3: Card Draw and Hand Update

The Cult draws cards based on draw bonuses, then updates hand composition.

*Definition 7* (Draw Count). Given garden configuration  $\mathbf{G}'$ , the number of cards drawn is:

$$n(\mathbf{G}') = 1 + \text{Bonus}(\mathbf{G}') = 1 + \sum_s \mathbb{1}[G'_s \geq 2] \quad (9)$$

*Definition 8* (Hand Transition). Let  $\mathbf{Y} = (Y_F, Y_R, Y_M, Y_B)$  be the cards drawn, where  $Y_s$  is the count of suit- $s$  cards. Under a fresh-deck approximation,  $\mathbf{Y}$  follows a multivariate hypergeometric distribution:

$$\mathbf{Y} \sim \text{MultiHypergeom}(54; 12, 12, 12, 18; n) \quad (10)$$

After optimal discard, the new hand indicator is:

$$h''_s = \mathbb{1}[H_s = 1 \text{ or } Y_s \geq 1] \quad (11)$$

Since we only need to know whether  $Y_s \geq 1$ , the transition probability simplifies to:

$$\mathbb{P}(h''_s = 1 \mid H_s, n) = \begin{cases} 1 & \text{if } H_s = 1 \\ q(n) & \text{if } H_s = 0 \end{cases} \quad (12)$$

where  $q(n)$  is the hit probability (see Proposition 19). Due to the fresh-deck approximation, drawing the suits are conditionally independent:

$$\mathbb{P}(\mathbf{h}'' \mid \mathbf{h}', \mathbf{G}') = \prod_s \mathbb{P}(h''_s \mid H_s, n(\mathbf{G}')) \quad (13)$$

### 2.2.4 Phase 4: Attrition

Opponents destroy gardens through combat and special abilities; again, as this depends so heavily on factors outside of player control, we introduce a parameter to simulate interference. Also, each garden destroyed forces the Cult to discard one card at random from their hand.

*Definition 9* (Attrition Model). Total garden losses in a turn follow  $L \sim \text{Poisson}(\lambda)$ , where  $\lambda > 0$  is the *interference rate*. We assume that losses are allocated uniformly at random among existing gardens. Given total losses  $L = \ell$  and garden configuration  $\mathbf{G}'$  with  $|\mathbf{G}'| > 0$ , the per-suit losses  $(L_F, L_R, L_M)$  follow:

$$(L_F, L_R, L_M) \mid L = \ell, \mathbf{G}' \sim \text{Multinomial} \left( \ell; \frac{G'_F}{|\mathbf{G}'|}, \frac{G'_R}{|\mathbf{G}'|}, \frac{G'_M}{|\mathbf{G}'|} \right)$$

The updated garden counts are:

$$G''_s = \max(G'_s - L_s, 0) \quad (14)$$

*Assumption 10* (Random Forced Discard). Each destroyed garden forces the Cult to discard a card randomly (after the Cult has already discarded down to no more than the hand limit  $H = 5$ ). With  $D = \sum_s L_s$  gardens destroyed, the Cult discards  $D$  cards sequentially from a shrinking hand of sizes  $H, H-1, \dots$  which is equivalent to choosing  $D$  cards uniformly without replacement from  $H$  cards. In the binary hand model, suit  $s$  is eliminated from hand iff all  $c_s$  copies of suit- $s$  cards are among the  $D$  discards:

$$\mathbb{P}(\text{eliminate } s \mid c_s, H, D) = \frac{\binom{H-c_s}{D-c_s}}{\binom{H}{D}} \cdot \mathbb{1}[D \geq c_s] \quad (15)$$

Since  $c_s$  is unobserved in the binary model, we average over the conditional distribution  $c_s \mid c_s \geq 1$ . Under the fresh-deck approximation, the hand of  $H = 5$  is treated as a random sample from the 54-card deck, so  $c_s \sim \text{Hypergeometric}(54, 12, 5)$ . Let  $\omega_c = \mathbb{P}(c_s = c \mid c_s \geq 1)$  be the conditional weights:

| $c$        | 1     | 2     | 3     | $\geq 4$ |
|------------|-------|-------|-------|----------|
| $\omega_c$ | 0.581 | 0.328 | 0.082 | 0.009    |

The suit elimination probability for  $D$  discards is then:

$$\gamma(D) = \mathbb{P}(h'''_s = 0 \mid h''_s = 1, D) = \sum_{c=1}^{\min(H, D)} \omega_c \frac{\binom{H-c}{D-c}}{\binom{H}{D}}$$

with the post-attrition hand survival:

$$\mathbb{P}(h'''_s = 1 \mid h''_s = 1, D) = 1 - \gamma(D) \quad (16)$$

and  $\mathbb{P}(h'''_s = 1 \mid h''_s = 0) = 0$ .

Precomputed values of  $\gamma$  for  $H = 5$ :

| Gardens lost $D$          | 1     | 2     | 3     |
|---------------------------|-------|-------|-------|
| $\gamma(D)$ (elim. prob.) | 0.116 | 0.265 | 0.455 |

At typical attrition rates ( $\lambda = 0.2$ ),  $\mathbb{P}(D \geq 1) \approx 0.18$  and  $\mathbb{P}(D \geq 2) \approx 0.02$ , so forced discards affect hand composition only occasionally.

**Remark 11** (Interpretation of the rate  $\lambda$ ). The parameter  $\lambda$  is the expected number of gardens destroyed per turn by opponents. In passive games,  $\lambda \approx 0.1$ – $0.2$ ; in aggressive games where opponents target the Cult,  $\lambda \approx 0.4$ – $0.6$ . In most games,  $\lambda$  might vary throughout the game, and soar if the Cult is in a position to win.

**Proposition 12** (Attrition Transition Probabilities). *For a single suit with  $g$  gardens (out of  $|\mathbf{G}|$  total), the probability of losing exactly  $k$  gardens is given by the Poisson-Multinomial compound:*

$$\mathbb{P}(L_s = k \mid \mathbf{G}') = \sum_{\ell=k}^{\infty} \frac{e^{-\lambda} \lambda^{\ell}}{\ell!} \binom{\ell}{k} \left( \frac{g}{|\mathbf{G}'|} \right)^k \left( 1 - \frac{g}{|\mathbf{G}'|} \right)^{\ell-k}$$

(for computational purposes, we truncate at  $\ell = 10$  since  $\mathbb{P}(L > 10) \approx 0$ .)

In particular, the probability of losing zero gardens in suit  $s$  ('survival probability') is:

$$\mathbb{P}(L_s = 0 \mid \mathbf{G}') = e^{-\lambda g/|\mathbf{G}'|} = p_{\text{surv}} \quad (17)$$

*Proof.* By the Poisson thinning property, if  $L \sim \text{Poisson}(\lambda)$  and each loss hits suit  $s$  independently with probability  $p = g/|\mathbf{G}'|$ , then  $L_s \sim \text{Poisson}(\lambda p)$ , which implies  $\mathbb{P}(L_s = 0) = e^{-\lambda p}$  [2].  $\square$

### 2.3 Full Transition Kernel

**Theorem 13** (Transition Kernel). *Under threshold policy  $\pi_{\theta}$ , the transition probability from state  $\mathbf{x} = (\mathbf{G}, V, \mathbf{h})$  to state  $\mathbf{x}' = (\mathbf{G}', V', \mathbf{h}''')$  is:*

$$\mathbb{P}_{\theta}(\mathbf{x}' \mid \mathbf{x}) = \sum_{\mathbf{G}', \mathbf{h}', \mathbf{h}''} \mathbb{P}_{\text{grow}}(\mathbf{G}' \mid \mathbf{G}) \cdot \mathbb{P}_{\text{draw}}(\mathbf{h}' \mid \mathbf{h}', \mathbf{G}') \cdot \mathbb{1}[\text{score}(\mathbf{G}', \mathbf{h}', V; \theta) = (V', \mathbf{h}'')] \cdot \mathbb{P}_{\text{attrit}}(\mathbf{G}'', \mathbf{h}''' \mid \mathbf{G}', \mathbf{h}'')$$

where  $\text{score}(\cdot)$  is the deterministic scoring function from Definition 6, and  $\mathbb{P}_{\text{attrit}}$  includes the random forced discard from Assumption 10:

$$\mathbb{P}_{\text{attrit}}(\mathbf{G}'', \mathbf{h}''' \mid \mathbf{G}', \mathbf{h}'') = \mathbb{P}_{\text{attrit}}(\mathbf{G}'' \mid \mathbf{G}') \cdot \prod_s \mathbb{P}(h_s''' \mid h_s'', D)$$

with  $D = \sum_s (G_s' - G_s'')$  and per-suit hand survival given above.

**Definition 14** (Transition Matrix). The transition matrix  $\mathbf{P}_{\theta}$  is a  $(|\mathcal{S}| + 1) \times (|\mathcal{S}| + 1)$  matrix with entries  $[\mathbf{P}_{\theta}]_{\mathbf{x}, \mathbf{x}'} = \mathbb{P}_{\theta}(\mathbf{x}' \mid \mathbf{x})$ . States with  $V' \geq 30$  transition to the absorbing victory state with probability 1. Let:

$$\mathbf{P}_{\theta} = \begin{pmatrix} \mathbf{Q}_{\theta} & \mathbf{r}_{\theta} \\ \mathbf{0}^{\top} & 1 \end{pmatrix} \quad (18)$$

where  $\mathbf{Q}_{\theta}$  is the  $|\mathcal{S}| \times |\mathcal{S}|$  submatrix of transient-to-transient transitions, and  $\mathbf{r}_{\theta} \in [0, 1]^{|\mathcal{S}|}$  is the absorption vector whose  $\mathbf{x}$ -th entry gives the probability of transitioning from transient state  $\mathbf{x}$  directly to the absorbing victory state in one step:

$$[\mathbf{r}_{\theta}]_{\mathbf{x}} = \mathbb{P}_{\theta}(V' \geq 30 \mid \mathbf{X}_t = \mathbf{x}). \quad (19)$$

Note that absorption can only occur during the scoring phase (Phase 2); the subsequent draw and attrition phases cannot increase VP.

### 2.4 Victory Time Analysis

**Definition 15** (Victory Time). The victory time under policy  $\pi_{\theta}$  starting from state  $\mathbf{x}$  is:

$$\tau_{\theta}(\mathbf{x}) = \inf\{t \geq 0 : V_t \geq 30 \mid \mathbf{X}_0 = \mathbf{x}\} \quad (20)$$

**Proposition 16** (Hitting Time Moments). *Let  $\mathbf{N}_{\theta} = (\mathbf{I} - \mathbf{Q}_{\theta})^{-1}$  be the fundamental matrix. Then:*

1. The expected victory time from state  $\mathbf{x}$  is:

$$\mathbb{E}[\tau_{\theta}(\mathbf{x})] = [\mathbf{N}_{\theta} \cdot \mathbf{1}]_{\mathbf{x}} \quad (21)$$

2. The variance is:

$$\text{Var}(\tau_{\theta}(\mathbf{x})) = [(2\mathbf{N}_{\theta} - \mathbf{I}) \cdot \mathbf{t}_{\theta}]_{\mathbf{x}} - [\mathbf{t}_{\theta}]_{\mathbf{x}}^2 \quad (22)$$

for  $\mathbf{t}_{\theta} = \mathbf{N}_{\theta} \cdot \mathbf{1}$  the expected hitting time vector.

3. The probability of victory in exactly  $k$  turns is:

$$\mathbb{P}(\tau_{\theta}(\mathbf{x}) = k) = [\mathbf{Q}_{\theta}^{k-1} \cdot \mathbf{r}_{\theta}]_{\mathbf{x}} \quad (23)$$

*Proof.* Standard for absorbing Markov chains [3].  $\square$

**Definition 17** (Initial State). The Cult begins with one garden in a chosen clearing. Wlog. we fix this as Fox, giving  $\mathbf{G}_0 = (1, 0, 0)$  and  $V_0 = 0$ . The Cult also draws 3 cards from the full 54-card deck, so the initial hand state  $\mathbf{h}_0$  is random. Under the fresh-deck approximation, the initial distribution is:

$$\pi_0(\mathbf{h}) = \prod_{s \in \{F, R, M\}} q(3)^{h_s} (1 - q(3))^{1-h_s} \quad (24)$$

where  $q(3) \approx 0.537$  is the probability of drawing at least one card of a given suit in 3 draws (c.f. Proposition 19). The expected victory time from this initial distribution is:

$$\mathbb{E}[\tau_{\theta}] = \sum_{\mathbf{h} \in \{0,1\}^3} \pi_0(\mathbf{h}) \cdot [\mathbf{t}_{\theta}]_{((1,0,0),0,\mathbf{h})} \quad (25)$$

where  $\mathbf{t}_{\theta} = (\mathbf{I} - \mathbf{Q}_{\theta})^{-1} \mathbf{1}$  is the hitting time vector.

The transition matrix  $\mathbf{P}_{\theta}$  is sparse and can be easily constructed; the algorithm is in (Appendix A).

### 3 CARD DRAW ANALYSIS

#### 3.1 The Hypergeometric Distribution

The deck contains 54 cards: 12 Fox, 12 Rabbit, 12 Mouse, 18 Bird. The Cult's "Hatred of Birds" means only 36 cards (two-thirds) are usable for scoring.

**Proposition 18** (Draw Distribution). *Drawing  $n$  cards, the number of suit- $s$  cards follows:*

$$X_s \sim \text{Hypergeometric}(54, 12, n) \quad (26)$$

with mean  $\mathbb{E}[X_s] = 12n/54 = 2n/9$ .

*Proof.* This is a standard hypergeometric distribution with parameters which is mathematically equivalent to drawing cards from a deck without replacement [4]. The mean follows from linearity of expectation.  $\square$

**Proposition 19** (Hit Probability). *The probability of drawing at least one card of suit  $s$  is:*

$$q(n) = \mathbb{P}(X_s \geq 1) = 1 - \frac{\binom{42}{n}}{\binom{54}{n}} \quad (27)$$

Since the Cult can score multiple suits per turn, the expected VP per turn depends not just on the per-suit hit probability  $q(n)$  but on how many distinct suits appear in a single draw. We therefore consider  $K = |\{s : X_s \geq 1\}|$ , the number of scorable suits.

| Cards drawn $n$                   | 1     | 2     | 3     | 4     |
|-----------------------------------|-------|-------|-------|-------|
| $q(n)$                            | 0.222 | 0.399 | 0.540 | 0.652 |
| $\mathbb{E}[\text{suited cards}]$ | 0.67  | 1.33  | 2.00  | 2.67  |
| $\mathbb{E}[K]$                   | 0.67  | 1.20  | 1.62  | 2.20  |

*Proof.* See (Appendix B).  $\square$

**Proposition 20** (Multi-Suit Availability). *Let  $K = |\{s : X_s \geq 1\}|$  be the number of suits with at least one available card. For  $n = 4$  draws from a fresh deck:*

| $K \geq$    | 1     | 2     | 3     |
|-------------|-------|-------|-------|
| Probability | 0.990 | 0.759 | 0.418 |

In particular, with full draw bonuses ( $n = 4$ ), the Cult has 76% chance of scoring 2+ suits per turn.

### 4 OPTIMAL THRESHOLD DERIVATION

We prove that  $\theta^* = 2$  is optimal via two approaches: delay value analysis and Markov chain computation.

#### 4.1 Delay Value Characterization

We calculate the expected VP change from waiting an additional turn before scoring a suit.

**Definition 21** (Delay Value). For a suit with  $g$  gardens, the delay value is:

$$\Delta(g) = \mathbb{E}[\text{VP delay-then-score}] - \mathbb{E}[\text{VP score-now}]$$

Immediate scoring is optimal when  $\Delta(g) \leq 0$ .

**Proposition 22** (Delay Value Formula). *Let  $\mu$  be the growth probability and  $p_\ell = 1 - e^{-\lambda/|G|}$  be the per-suit loss probability ( $= 1 - e^{-\lambda/3}$  under diversified play). If we delay one turn, the garden count evolves to  $g' \in \{g-1, g, g+1\}$ : with probability  $\mu(1-p_\ell)$  the garden grows and survives, with probability  $(1-\mu)(1-p_\ell)$  nothing changes, and with probability  $p_\ell$  a garden is lost. (assuming that attrition removes at most one garden per suit, valid for small  $\lambda$ .)*

$$\begin{aligned} \mathbb{E}[V_{\text{delay}}] &= \mu(1-p_\ell)S(g+1) + (1-\mu)(1-p_\ell)S(g) + p_\ell S(g-1) \\ \implies \Delta(g) &= \mu(1-p_\ell)[S(g+1) - S(g)] - p_\ell[S(g) - S(g-1)] \end{aligned}$$

Evaluating at each garden count with parameters  $\mu = 0.4$ ,  $p_\ell = 0.065$  (corresp. to  $\mu = 0.4$ ,  $\lambda = 0.2$ ):

| $g$         | 1     | 2     | 3     | 4     | 5      |
|-------------|-------|-------|-------|-------|--------|
| $S(g)$      | 0     | 2     | 2     | 3     | 4      |
| Gain        | 2     | 0     | 1     | 1     | —      |
| Loss        | 0     | 2     | 0     | 1     | 1      |
| $\Delta(g)$ | +0.75 | -0.13 | +0.37 | +0.31 | -0.065 |

Two features stand out. At  $g = 1$ , scoring yields 0 VP, so there is no opportunity cost to waiting;  $\Delta(1) > 0$  for all parameter values. At  $g = 2$ , the scoring plateau  $S(2) = S(3) = 2$  means growth provides zero upside, but attrition to  $g = 1$  destroys all scoring value (downside = 2). In particular,  $\Delta(2) = -2p_\ell < 0 \forall \mu$ .

#### 4.2 Proof of Optimal Threshold

We show that  $\theta^* = 2$  by considering the build-up time.

**Theorem 23** (Optimal Threshold). *For  $\mu \in [0.2, 0.6]$  and  $\lambda \in [0.1, 0.5]$ , the optimal threshold is  $\theta^* = 2$ .*

*Proof.* **Claim 1** ( $\theta = 1$  is dominated). Since  $S(1) = 0$ , a policy that scores at  $g = 1$  earns zero VP from any suit with exactly one garden. Since  $S(g) \geq 2$  for all  $g \geq 2$ , any threshold  $\theta \geq 2$  dominates  $\theta = 1$  always.

**Claim 2** (Immediate scoring is optimal at  $g = 2$ ). As shown in the table above,  $\Delta(2) = -2p_\ell < 0$  for all  $\lambda > 0$ . This reflects the structure of  $S$ : the plateau  $S(2) = S(3) = 2$  means growth from  $g = 2$  to  $g = 3$  provides zero VP, but attrition from  $g = 2$  to  $g = 1$  causes a total loss of scoring value.

**Claim 3** ( $\theta \geq 3$  is suboptimal). Although  $\Delta(3)$  and  $\Delta(4)$  are positive, a threshold policy must first build to that level before scoring begins.

For  $\theta = 3$ , the case is immediate: since  $S(3) = S(2)$  and the draw bonus activates at  $g = 2$ , the scoring rate is identical to  $\theta = 2$ 's, and  $\theta = 3$  requires strictly more build-up time so  $\mathbb{E}[\tau(3)] > \mathbb{E}[\tau(2)] \forall \mu, \lambda$ .

For  $\theta \geq 4$ , we use a decomposition (Appendix C):  $\mathbb{E}[\tau(\theta)] \approx T_{\text{build}}(\theta) + 30/\bar{V}(\theta)$ , where  $\bar{V}(\theta) = S(\theta) \cdot \mathbb{E}[K | n(\theta)] \cdot p_{\text{surv}}$  is the expected VP per scoring turn.

Since all thresholds  $\theta \geq 2$  yield  $n(\theta) = 4$  (three suits above the draw bonus threshold),  $\mathbb{E}[K | n]$  and  $p_{\text{surv}}$  are common factors. The saving from raising the threshold to  $\theta$  is therefore:

$$\Delta_{\text{score}}(\theta) = \frac{30}{S(2) \cdot C} - \frac{30}{S(\theta) \cdot C} = \frac{30}{C} \cdot \frac{S(\theta) - S(2)}{S(2) \cdot S(\theta)}$$

where  $C = \mathbb{E}[K | n=4] \cdot p_{\text{surv}}$  is independent of  $\theta$ . Since  $C \leq 3$  (at most 3 suits scored) and  $S(\theta) \leq 4$ :

$$\Delta_{\text{score}}(\theta) \leq \frac{30}{C} \cdot \frac{S(\theta) - 2}{2 \cdot S(\theta)} \leq \frac{30}{C} \cdot \frac{1}{4} = \frac{7.5}{C} \quad (28)$$

using  $S(5) = 4$ , the largest available value. Since  $C \geq 1$ , this gives  $\Delta_{\text{score}} \leq 7.5$  turns saved. Meanwhile, the additional build-up cost satisfies:

$$\Delta_{\text{build}}(\theta) = T_{\text{build}}(\theta) - T_{\text{build}}(2) \geq \frac{\theta - 2}{\mu - \lambda/3} \quad (29)$$

since the net per-suit drift is at most  $\mu - \lambda/3$ , and all three suits must reach  $\theta$ . For the parameter range  $\mu \in [0.2, 0.6]$  and  $\lambda \in [0.1, 0.5]$ , the net drift  $\mu - \lambda/3$  ranges from 0.03 to 0.57, giving:

| $\theta$ | $\Delta_{\text{score}} \leq$                 | $\Delta_{\text{build}} \geq$ | Net  |
|----------|----------------------------------------------|------------------------------|------|
| 4        | $\frac{30}{C} \cdot \frac{1}{6} \approx 2.5$ | 3.5 to 67                    | loss |
| 5        | $\frac{30}{C} \cdot \frac{1}{4} \approx 3.8$ | 5.3 to 100                   | loss |

(with the most favorable value  $C = 2$ ). Under the most favorable reasonable drift ( $\mu = 0.6, \lambda = 0.1$ ),  $\Delta_{\text{build}}$  exceeds  $\Delta_{\text{score}}$  for  $\theta \geq 4$ , so in the viable regime  $\mu > \lambda/3$ , the build penalty dominates.  $\square$

### 4.3 Endgame Adjustment

**Proposition 24** (Endgame Threshold). *When  $V + S(G_s) \geq 30$ , the optimal action is to score immediately regardless of  $\theta$ .*

**Corollary 25** (Victory-Aware Policy). *The optimal threshold is state-dependent:*

$$\theta^*(V) = \min(\theta_{\text{base}}^*, g^* \text{ s.t. } S(g^*) \geq 30 - V) \quad (30)$$

### 4.4 Exact Markov Chain Computation

The transition matrix  $\mathbf{P}_\theta$  defined in Section 2 enables exact computation of  $\mathbb{E}[\tau(\theta)]$ .

**Theorem 26** (Exact Victory Times). *Let  $\pi_0$  be the initial distribution from Definition 17, with  $\mathbf{G}_0 = (1, 0, 0)$  and  $\mathbf{h}_0$  drawn from 3 initial cards. The fundamental matrix  $\mathbf{N}_\theta = (\mathbf{I} - \mathbf{Q}_\theta)^{-1}$  yields the hitting time vector  $\mathbf{t}_\theta = \mathbf{N}_\theta \cdot \mathbf{1}$ , and the expected victory time is  $\mathbb{E}[\tau_\theta] = \pi_0 \cdot \mathbf{t}_\theta$  (equation (25)). Solving the sparse linear system  $(\mathbf{I} - \mathbf{Q}_\theta)\mathbf{t} = \mathbf{1}$  gives exact values for these quantities.*

We do not provide an explicit proof beyond that this is merely a conclusion of what we constructed, but we do provide a simulation, and the algorithm (Appendix D).

| $\theta$                  | 1        | 2     | 3     | 4     | 5     |
|---------------------------|----------|-------|-------|-------|-------|
| $\mathbb{E}[\tau_\theta]$ | $\infty$ | 10.95 | 12.47 | 13.85 | 16.03 |
| $\text{Var}(\tau_\theta)$ | —        | 6.68  | 9.04  | 11.31 | 14.91 |

Parameters:  $\mu = 0.4, \lambda = 0.2$ . Expectations taken over both the initial hand draw and the trajectory.

## 5 DIVERSIFICATION ANALYSIS

The threshold policy  $\pi_\theta$  prescribes *when* to score, but the Cult must also decide *how to build*: should gardens be spread evenly across suits or concentrated into one? The Markov chain model provides an answer.

Under  $\theta^* = 2$ , a garden at  $g = 1$  contributes nothing, and a garden at  $g = 3$  contributes no more than  $g = 2$  since  $S(3) = S(2)$ . The strategy question is therefore whether to activate all three scoring tracks at  $g = 2$  or to push a suit higher at the expense of another.

**Proposition 27** (Diversification Dominance). *Let  $\mathbf{G}_D = (2, 2, 2)$  and  $\mathbf{G}_C = (4, 2, 0)$  be two configurations with 6 total gardens. Then*

$$\mathbb{E}[V_{\text{turn}} | \mathbf{G}_D] > \mathbb{E}[V_{\text{turn}} | \mathbf{G}_C] \quad (31)$$

*Proof.* (Sketch). The configurations differ in two quantities from our model. First,  $\mathbf{G}_D$  activates all three draw bonuses ( $n = 4$ ), while  $\mathbf{G}_C$  activates only two ( $n = 3$ ). By Proposition 19, the per-suit hit probability increases from  $q(3) = 0.540$  to  $q(4) = 0.652$ .

Second,  $\mathbf{G}_D$  opens three scoring tracks (each worth  $S(2) = 2$ ), while  $\mathbf{G}_C$  opens only two (worth  $S(4) + S(2) = 3 + 2 = 5$  combined).

Furthermore, at  $\lambda = 0.2$ , the per-suit survival probability for the diversified strategy is

$p_{2, \text{surv}} = e^{-\lambda \frac{2}{6}} \approx 0.936$ , the same as the concentrated strategy in its 2-garden suit, but in the 4-garden suit it

is  $p_{4,\text{surv}} = e^{-\lambda \frac{4}{6}} \approx 0.875$ . Therefore:

$$\begin{aligned}\mathbb{E}[V_{\text{turn}} \mid \mathbf{G}_D] &= 3 \times S(2) \times q(4) \times p_{2,\text{surv}} = 3.66 \text{ VP/turn} \\ \mathbb{E}[V_{\text{turn}} \mid \mathbf{G}_C] &= [S(4)p_{4,\text{surv}} + S(2)p_{2,\text{surv}}] \times q(3) \\ &= 2.43 \text{ VP/turn}\end{aligned}$$

Notably, diversification yields a 50% higher rate.  $\square$

These results suggest a natural build sequence. In the early game, the Cult should prioritize reaching  $(2, 2, 2)$ , building one suit at a time:  $(2, 0, 0) \rightarrow (2, 2, 0) \rightarrow (2, 2, 2)$ , since each new suit at  $g = 2$  unlocks a draw bonus and scoring. Further gardens serve as attrition buffers but should not delay scoring: the delay value analysis (Section 4) shows  $\Delta(2) < 0$ , indicating that the Cult should score any suit at  $g \geq 2$  immediately until the endgame adjustment (Proposition 24) overrides the base threshold.

## 6 MONTE CARLO SIMULATION

We simulate  $N = 10,000$  games per configuration via an algorithm (Appendix D, Appendix E).

Table 1: Victory time by threshold (mean),  $\mu = 0.4$ .

| Policy      | $\lambda = 0.15$ |            | $\lambda = 0.30$ |            |
|-------------|------------------|------------|------------------|------------|
|             | Mean             | 95% CI     | Mean             | 95% CI     |
| Threshold-2 | <b>10.2</b>      | $\pm 0.04$ | <b>11.1</b>      | $\pm 0.05$ |
| Threshold-3 | 11.8             | $\pm 0.04$ | 13.0             | $\pm 0.06$ |
| Threshold-4 | 13.3             | $\pm 0.05$ | 14.8             | $\pm 0.08$ |

Table 2: Diversification comparison ( $\mu = 0.4$ ,  $\lambda = 0.2$ , Diversified =  $(2, 2, 2)$ , Concentrated =  $(4, 2, 2)$ ).

| Strategy     | Mean $T$ | VP/turn | $\mathbb{P}(\text{win by } t=10)$ |
|--------------|----------|---------|-----------------------------------|
| Diversified  | 10.0     | 3.1     | 0.67                              |
| Concentrated | 10.9     | 2.9     | 0.48                              |

In the simulation, the Concentrated strategy produces better results than expected (though still worse than Diversified). Inspecting trajectories shows that this is due to the two strategies often being ‘effectively’ equivalent due to the difficulty of maintaining the  $(2, 2, 2)$  position that allows constant full scoring.

Table 3: Sensitivity of  $\mathbb{E}[T]$  to parameters  $\xi$  (base  $\mu = 0.4$ ,  $\lambda = 0.2$ ,  $\theta = 2$ ).

| Parameter           | $\partial \mathbb{E}[T] / \partial \xi$ | Interpretation                |
|---------------------|-----------------------------------------|-------------------------------|
| Attrition $\lambda$ | +5.6                                    | 0.1 increase adds 0.56 turns  |
| Growth $\mu$        | -17.5                                   | 0.1 increase saves 1.75 turns |
| Threshold $\theta$  | +1.7                                    | 1 unit adds 1.7 turns         |

Growth rate is the dominant parameter, and the hardest to model, given that it depends so heavily on a player’s luck with Building and Conspiracies. In real play, higher  $\mu$  is likely to also yield higher  $\lambda$ , as the higher growth from Sanctify conspiracies is only possible if the Cult’s warriors are frequently attacked.

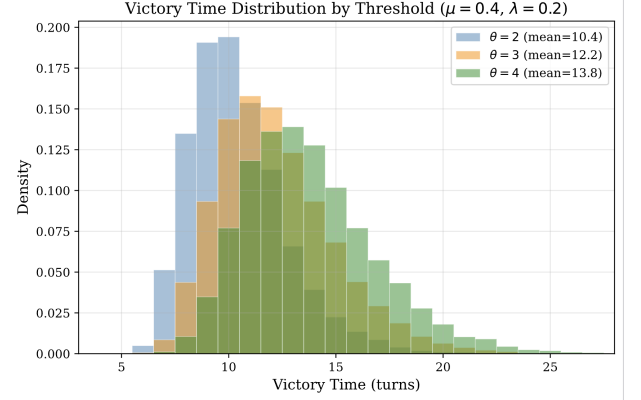


Figure 2: Distribution of values

The distribution is right-skewed with occasional long games ( $T > 15$ ) due to poor draws or heavy attrition.

## 7 CONCLUSION

By developing a Markov chain framework for Lizard Cult scoring in *Root*, we found:

1.  $\theta^* = 2$  is robustly optimal because it is simultaneously the minimum viable scoring level and the draw bonus activation threshold.
2. The one-turn delay value  $\Delta(g)$  is positive at  $g = 1$  and negative at  $g = 2$  (no upside, downside risk), which reinforces  $\theta^* = 2$ .
3. Multi-suit scoring with configuration  $(2, 2, 2)$  yields higher VP rates than blind growth.

## REFERENCES

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## 8 APPENDIX

The algorithms, some calculations, and more plots can be found here.

## A ALGORITHM FOR MATRIX COMPUTATION

---

**Algorithm 1:** Transition Matrix Construction
 

---

**Input:** Parameters  $\mu, \lambda$ , threshold  $\theta$   
**Output:** Sparse transition matrix  $\mathbf{P}_\theta$   
 Precompute  $\gamma(D)$  for  $D = 0, 1, \dots, 5$ ;  
 Initialize sparse matrix  $\mathbf{P}_\theta$  of size  $(|\mathcal{S}| + 1) \times (|\mathcal{S}| + 1)$ ;  
**for each state**  $\mathbf{x} = (\mathbf{G}, V, \mathbf{h}) \in \mathcal{S}$  **do**  
   **for each growth outcome**  $\mathbf{G}'$  **do**  
 Compute  $p_{\text{grow}} = P_{\text{grow}}(\mathbf{G}' | \mathbf{G})$  via Definition 4;  
**if**  $p_{\text{grow}} = 0$  **then**  
   **continue**  
**end**  
 $n \leftarrow n(\mathbf{G}')$ ;  
 $(V', \mathbf{h}') \leftarrow \text{score}(\mathbf{G}', \mathbf{h}, V; \theta)$ ;  
**if**  $V' \geq 30$  **then**  
    $\mathbf{P}_\theta[\mathbf{x}, \text{absorbing}] += p_{\text{grow}}$ ;  
**else**  
   **for each draw outcome**  
      $\mathbf{h}'' \in \{0, 1\}^3$  **do**  
       Compute  
        $p_{\text{draw}} = P_{\text{draw}}(\mathbf{h}'' | \mathbf{h}', n)$ ;  
       **if**  $p_{\text{draw}} = 0$  **then**  
         **continue**  
       **end**  
       **for each attrition outcome**  $\mathbf{G}''$   
         with  $G_s'' \leq G_s'$  **do**  
           Compute  
            $p_{\text{attrit}} = P_{\text{attrit}}(\mathbf{G}'' | \mathbf{G}')$ ;  
            $D \leftarrow \sum_s (G_s' - G_s'')$ ;  
           **for each hand outcome**  
              $\mathbf{h}''' \in \{0, 1\}^3$  **do**  
                $p_{\text{discard}} \leftarrow \prod_s \mathbb{P}(h_s''' | h_s'', D, \gamma)$ ;  
               **if**  $p_{\text{discard}} = 0$  **then**  
                 **continue**  
               **end**  
                $\mathbf{x}' \leftarrow (\mathbf{G}'', V', \mathbf{h}''')$ ;  
                $\mathbf{P}_\theta[\mathbf{x}, \mathbf{x}'] += p_{\text{grow}} \cdot p_{\text{draw}} \cdot p_{\text{attrit}} \cdot p_{\text{discard}}$ ;  
             **end**  
           **end**  
       **end**  
     **end**  
   **end**  
**end**  
 $\mathbf{P}_\theta[\text{absorbing}, \text{absorbing}] \leftarrow 1$ ;  
**return**  $\mathbf{P}_\theta$

---

## B HYPERGEOMETRIC COMPUTATIONS

*Proof of Proposition 19.* The probability of drawing zero cards of suit  $s$  in  $n$  draws:

$$\mathbb{P}(X_s = 0) = \frac{\binom{42}{n}}{\binom{54}{n}} \quad (32)$$

For  $n = 3$ :

$$\mathbb{P}(X_s = 0) = \frac{42 \cdot 41 \cdot 40}{54 \cdot 53 \cdot 52} = \frac{68880}{148824} = 0.463 \quad (33)$$

Thus  $q(3) = 1 - 0.463 = 0.537 \approx 0.540$ .  $\square$

We also compute  $\mathbb{P}(K \geq k)$  for  $n = 4$  draws directly.

**Setup.** Let  $A_s = \{X_s \geq 1\}$  be the event of drawing at least one card of suit  $s$ . We have:

$$\mathbb{P}(A_s^c) = \frac{\binom{42}{4}}{\binom{54}{4}} = \frac{111930}{316251} = 0.354 \quad (34)$$

**Computing**  $\mathbb{P}(K = 0)$  (all Birds):

$$\mathbb{P}(K = 0) = \mathbb{P}(A_F^c \cap A_R^c \cap A_M^c) = \frac{\binom{18}{4}}{\binom{54}{4}} = \frac{3060}{316251} = 0.010 \quad (35)$$

**Computing**  $\mathbb{P}(K \geq 1)$ :

$$\mathbb{P}(K \geq 1) = 1 - \mathbb{P}(K = 0) = 1 - 0.010 = 0.990 \quad (36)$$

**Computing**  $\mathbb{P}(K = 1)$  (exactly one suit): By symmetry and inclusion-exclusion,

$$\begin{aligned} \mathbb{P}(K = 1) &= 3 \cdot \mathbb{P}(A_F \cap A_R^c \cap A_M^c) \\ &= 3 \cdot \frac{\binom{30}{4} - \binom{18}{4}}{\binom{54}{4}} = 0.231 \end{aligned}$$

**Computing**  $\mathbb{P}(K \geq 2)$ :

$$\mathbb{P}(K \geq 2) = 1 - \mathbb{P}(K = 0) - \mathbb{P}(K = 1) = 1 - 0.010 - 0.231 = 0.759$$

## C TWO-PHASE DECOMPOSITION

We justify the decomposition underlying Theorem 23. Under threshold policy  $\pi_\theta$ , gameplay divides into:

1. **Build phase:** Accumulate gardens to level  $\theta$  in each suit (expected time  $T_{\text{build}}(\theta)$ )
2. **Score phase:** Score repeatedly until victory (expected time  $30/\bar{V}(\theta)$ )

The expected victory time is therefore:

$$\mathbb{E}[\tau(\theta)] \approx T_{\text{build}}(\theta) + \frac{30}{\bar{V}(\theta)} \quad (37)$$

In reality, the Cult may score opportunistically during the build phase, and attrition can knock gardens below threshold during the score phase. However, for threshold  $\theta = 2$ , the build phase is shorter and the score phase dominates, making this approximation accurate.



## D MONTE CARLO SIMULATION

**Algorithm 2:** Lizard Cult Simulation

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**Input:** Growth  $\mu$ , attrition  $\lambda$ , threshold  $\theta$ ,  $N$   
**Output:** Victory times  $\{T_i\}$

```

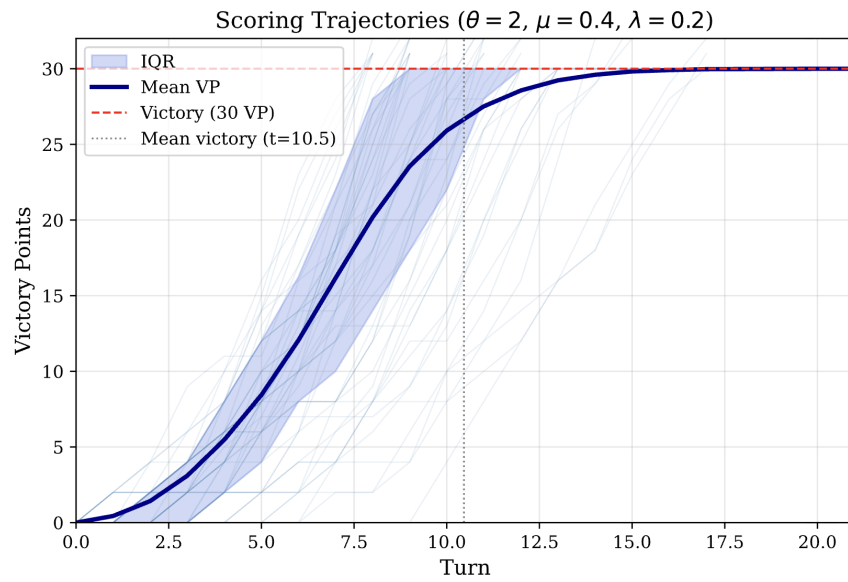
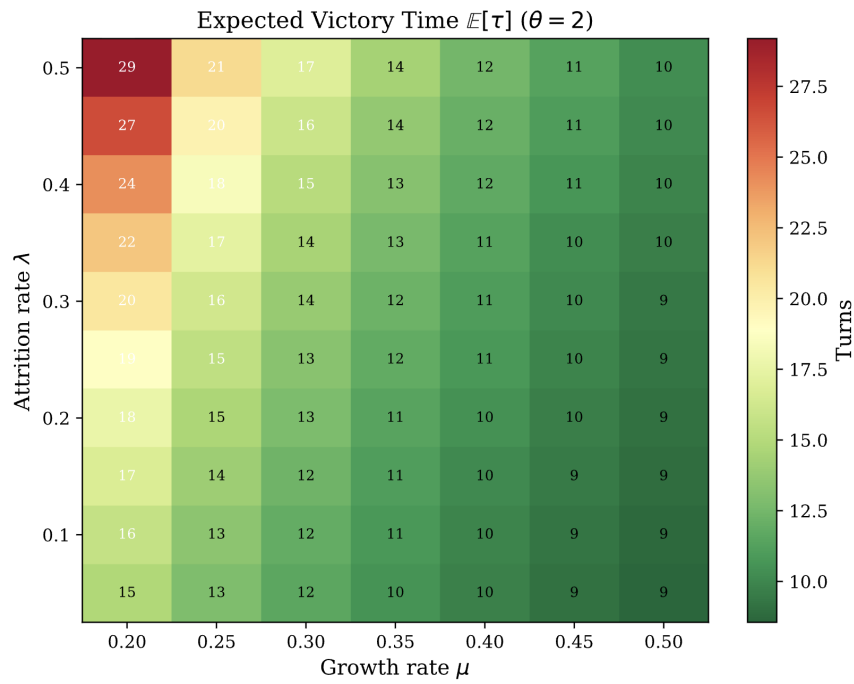
for  $i = 1$  to  $N$  do
   $\mathbf{G} \leftarrow (1, 0, 0)$ ,  $V \leftarrow 0$ ,  $t \leftarrow 0$ ;
  Draw  $\mathbf{H}$  from
    MultiHypergeom(54; 12, 12, 12, 18; 3);
  while  $|\mathbf{H}| > 5$  do
    Discard one Bird if possible, else
    random
  end
  while  $V < 30$  do
     $t \leftarrow t + 1$ ;
    // Phase 1: Garden growth
    for  $s \in \{F, R, M\}$  do
      if  $U \sim \text{Uniform}(0, 1) < \mu$  and
         $G_s < 5$  then
         $G_s \leftarrow G_s + 1$ 
      end
    end
    // Phase 2: Scoring
    for  $s \in \{F, R, M\}$  do
      if  $G_s \geq \theta$  and  $H_s \geq 1$  then
         $V \leftarrow V + S(G_s)$ ;  $H_s \leftarrow H_s - 1$ 
      end
    end
    if  $V \geq 30$  then
      break
    end
    // Phase 3: Card draw
     $n \leftarrow 1 + \sum_s \mathbb{1}[G_s \geq 2]$ ;
    Draw  $\Delta\mathbf{H}$  from
      MultiHypergeom(54; 12, 12, 12, 18;  $n$ );

     $\mathbf{H} \leftarrow \mathbf{H} + \Delta\mathbf{H}$ ;
    while  $|\mathbf{H}| > 5$  do
      Discard one Bird if possible, else
      random
    end
    // Phase 4: Attrition
     $L \sim \text{Poisson}(\lambda)$ ; allocate uniformly to
    get  $(L_F, L_R, L_M)$ ;
     $D \leftarrow \sum_s \min(L_s, G_s)$ ;
    for  $s \in \{F, R, M\}$  do
       $G_s \leftarrow \max(G_s - L_s, 0)$ 
    end
    for  $d = 1$  to  $D$  do
      if  $|\mathbf{H}| > 0$  then
        Remove one card uniformly at
        random from  $\mathbf{H}$ 
      end
    end
  end
   $T_i \leftarrow t$ ;
end

```

---

## E OTHER PLOTS AND TRAJECTORIES

Figure 3: Trajectories for scoring from Monte Carlo simulation,  $N = 10000$ Figure 4: Heatmap of sensitivity to parameters  $\mu, \lambda$



# The Lizard Cult





**Revenge**  
Your removed defending warriors become acolytes.



**Hatred of Birds**  
Bird cards are not wild for your rituals.



**Pilgrims**  
You rule any clearings where you have any gardens.

**Crafted Items**

*Vagabond can give you cards to take these items.*

**Birdsong**

- Adjust the Outcast.
- Discard Lost Souls.
- Perform Conspiracies in Outcast clearings.

**Daylight**

Reveal any number of cards from your hand, and perform one ritual for each.



**Build**  
Place a garden in a matching clearing you rule.



**Recruit**  
Place a warrior in a matching clearing.



**Score** once per suit per turn  
Discard an unrevealed card to score 1/2 of rightmost empty, matching Gardens space.



**Sacrifice**  
Place one warrior in the Acolytes box.

**Evening**

- Return revealed cards to hand.
- Craft using gardens matching Outcast suit.
- Draw 1 card, plus 1 card per  showing. Discard down to 5 cards.

**Conspiracies**

 **Crusade**  
Spend 2 acolytes to battle in any Outcast clearing or may move from any Outcast clearing and then may battle in the destination clearing.

 **Convert**  
Spend 2 acolytes to remove an enemy warrior in an Outcast clearing and place a warrior there.

 **Sanctify**  
Spend 3 acolytes to remove an enemy building in an Outcast clearing and place a garden there.

**Acolytes**

**Gardens**

*When a garden is removed from the map, discard a random card from your hand.*

| +0                                                                                  | +2                                                                                  | +3                                                                                  | +4                                                                                    |
|-------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------|
|    |    |    |    |
|   |  |  |  |
|  |  |  |  |

**The Outcast**

In Birdsong, most common suit of Lost Souls, ignoring birds, becomes Outcast. If already Outcast, flip to Hated Outcast.

*If Outcast suit is Hated, all conspiracies cost one less acolyte.*





**Lost Souls**

*All spent and discarded cards go here.*



Figure 5: The game board for the Lizard Cult as seen in *Root* [1]